Declarations

Reconstruction Order, Not Solver Family, Governs Control Accuracy: A Comparison of Neural and Classical HJB Solvers Against an Exact Reference

CRediT authorship contribution statement

Jingwen Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Fei Ma**: Conceptualization, Supervision, Methodology, Writing – review & editing, Project administration.

To be confirmed by both authors before submission: the corresponding author is not yet marked here, and funding acquisition and resources have been left out deliberately rather than assumed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code and derived results. All code, the preprocessing manifest, the temporal data splits and every raw result file behind every table and figure are released (see the manuscript's Reproducibility section). The repository is archived with a DOI at the time of publication, and the archived version is the one the paper's numbers were produced with.

Market data. The minute-bar data used for the market calibration are vendor data and cannot be redistributed. The repository instead contains the full preprocessing that consumes them, the temporal split dates, and the derived quantities the calibration depends on (per-instrument volatility and the Roll effective spread), so the calibration tables can be re-checked without the bars themselves.

Environment. The neural results were produced with the pinned environment in `requirements.txt`; the grid results need no GPU. The verification suite runs on CPU and on a clean checkout with no market data present.

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To be completed by the authors. The compiler used for the timing comparison is numba, an open-source project; the reference baseline for the regression method is the published least-squares Monte Carlo line, cited in the manuscript.

Use of generative AI

To be confirmed by the authors, and stated honestly in whatever form is true. Elsevier requires disclosure of generative-AI use in the writing process. The measurements, code and verifications

in this work were produced and checked computationally; any assistance used in drafting must be declared here in the wording the publisher requires.